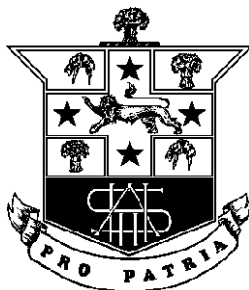


HURLSTONE AGRICULTURAL HIGH SCHOOL



MATHEMATICS – EXTENSION TWO

2010 HSC

ASSESSMENT TASK 1

Examiner ~ G Rawson

GENERAL INSTRUCTIONS

- | | |
|---|---|
| <ul style="list-style-type: none">• Reading Time – 3 minutes.• Working Time – 40 MINUTES.• Attempt all questions.• All necessary working should be shown in every question.• This paper contains two (2) questions. | <ul style="list-style-type: none">• Marks may not be awarded for careless or badly arranged work.• Board approved calculators may be used.• Each question is to be started on a new piece of paper.• This examination paper must NOT be removed from the examination room. |
|---|---|

STUDENT NAME: _____

TEACHER: _____

QUESTION ONE **15 marks** *Start a SEPARATE sheet*

(a) The ellipse $\mathcal{E}$ has equation $\frac{x^2}{25} + \frac{y^2}{16} = 1$.

(i) State the intercepts with the axes. **1**

(ii) Determine the eccentricity of $\mathcal{E}$ **1**

(iii) State the coordinates of the two foci. **1**

(iv) Write down the equation of the auxiliary circle. **1**

(b) (i) Sketch the function $f(x) = x^2 - c^2$, where $|c| > 1$, clearly showing its vertex and intercepts **1**

(ii) Hence, without using calculus, draw separate sketches, at least $\frac{1}{3}$ of a page, for each of the following curves.
For each sketch, show the original function with a dotted line, and clearly indicate turning points.

(A) $y = |f(x)|$ **2**

(B) $y = \frac{1}{f(x)}$ **2**

(C) $y = \sqrt{f(x)}$ **2**

(D) $y = [f(x)]^2$ **2**

(E) $y = [f(x)]^3$ **2**

QUESTION TWO **15 marks** *Start a SEPARATE sheet*

- (a) The points $P\left(cp, \frac{c}{p}\right)$ and $Q\left(cq, \frac{c}{q}\right)$ are points on the rectangular hyperbola $xy = c^2$.

Tangents to the rectangular hyperbola at P and Q intersect at the point R .

- (i) Show that the tangent to the rectangular hyperbola at any point $T\left(ct, \frac{c}{t}\right)$ has equation $x + t^2y - 2ct = 0$. **1**

- (ii) Find the coordinates of R . **2**

- (iii) If P and Q are variable points on the rectangular hyperbola which move so that $p^2 + q^2 = 2$, show that the equation of the locus of R is given by $xy + y^2 = 2c^2$. **3**

- (b) The points $P(a \sec \theta, b \tan \theta)$ and $Q(a \sec \phi, b \tan \phi)$ lie on the same branch of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, and PQ is a focal chord, passing through $S(ae, 0)$.

Use the gradients of PS and QS to show that $e = \frac{\sin \theta - \sin \phi}{\sin(\theta - \phi)}$. **4**

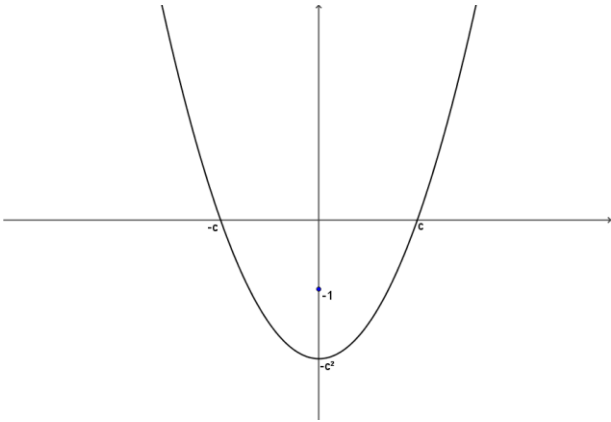
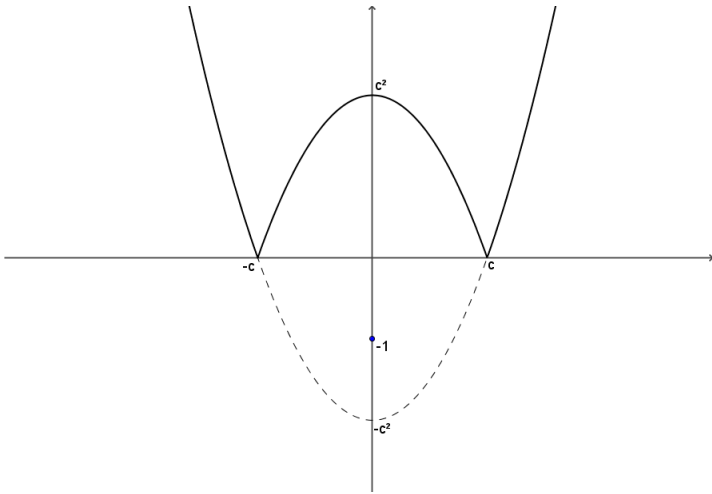
- (c) (i) Determine the real values of λ for which the equation

$$\frac{x^2}{4 - \lambda} + \frac{y^2}{2 - \lambda} = 1 \text{ defines}$$

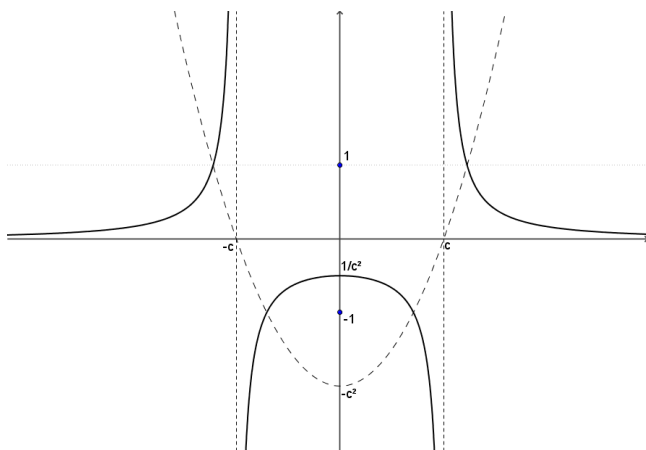
- (A) an ellipse **1**

- (B) a hyperbola **1**

- (ii) Describe how the shape of this curve changes as λ increases from 1 towards 2. What is the limiting position of the curve as λ approaches 2? **3**

Year 12	Mathematics Extension 2	Ass Task 1 2010 HSC
Question No. 1	Solutions and Marking Guidelines	
Outcomes Addressed in this Question		
E4 - uses efficient techniques for the algebraic manipulation required in dealing with questions such as those involving conic sections E6 - combines the ideas of algebra and calculus to determine the important features of the graphs of a wide variety of functions		
Part / Outcome	Solutions	Marking Guidelines
(a) E4	(i) (5, 0), (−5, 0), (0, 4), (0, −4)	1 mark : correct solution
	(ii) $b^2 = a^2(1 - e^2)$ $16 = 25(1 - e^2)$ $e^2 = \frac{9}{25}$ so $e = \frac{3}{5}$	1 mark : correct solution
	(iii) Foci: $(\pm ae, 0)$ ie (3,0) and (−3,0)	1 mark : correct solution
	(iv) auxiliary circle: $x^2 + y^2 = 25$	1 mark : correct solution
(b) (i) E6		1 mark : correct graph with correct intercepts shown
(ii)(A) E6		2 marks : correct graph with correct intercepts shown 1 mark : partially correct

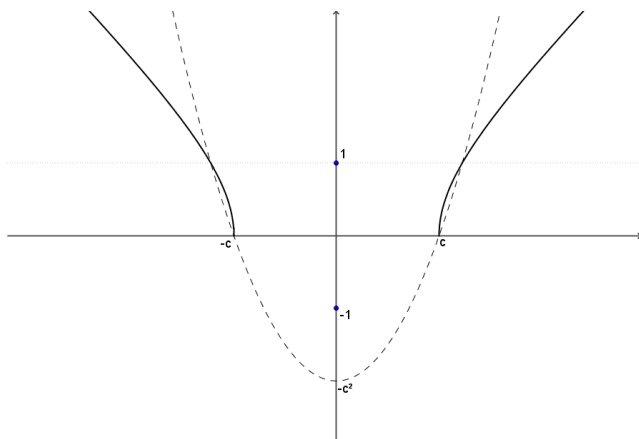
(ii)(B)
E6



2 marks : correct graph with correct intercepts shown

1 mark : partially correct

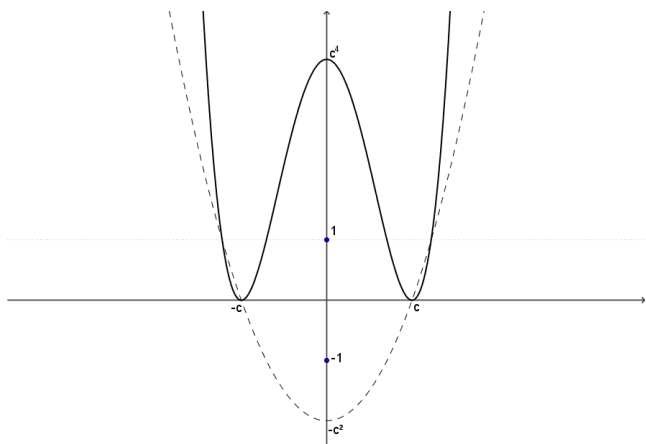
(ii)(C)
E6



2 marks : correct graph with correct intercepts shown

1 mark : partially correct

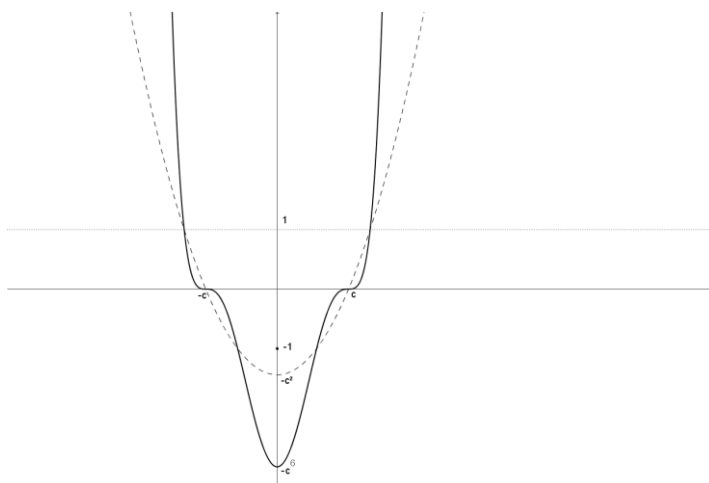
(ii)(D)
E6



2 marks : correct graph with correct intercepts shown

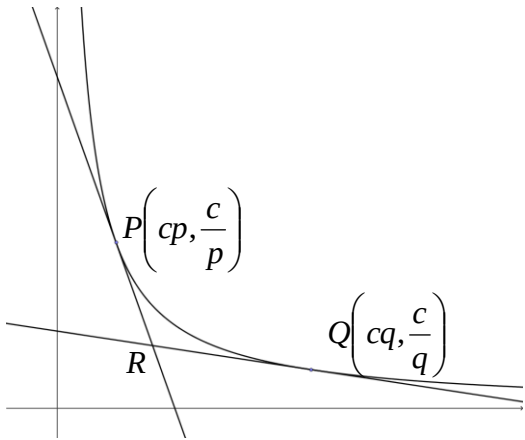
1 mark : partially correct

(ii)(E)
E6



2 marks : correct graph with correct intercepts shown

1 mark : partially correct

Year 12	Mathematics Extension 2	Ass Task 1 2010 HSC
Question No. 2	Solutions and Marking Guidelines	
Outcomes Addressed in this Question		
E4 - uses efficient techniques for the algebraic manipulation required in dealing with questions such as those involving conic sections		
Part / Outcome	Solutions	Marking Guidelines
(a)		
(a)(i)	$xy = c^2$ $y = \frac{c^2}{x}$ $y' = -\frac{c^2}{x^2}$ $m = -\frac{1}{t^2} \text{ at } x = ct$ $\therefore \text{ tangent is}$ $y - \frac{c}{t} = -\frac{1}{t^2}(x - ct)$ $t^2 y - ct = -x + ct$ $x + t^2 y - 2ct = 0$	1 mark : correct solution
(a)(ii)	tangent at P : $x + p^2 y - 2cp = 0 \dots(1)$ tangent at Q : $x + q^2 y - 2cq = 0 \dots(2)$ (1)-(2) $(p^2 - q^2)y = 2c(p - q)$ $y = \frac{2c}{p + q} \quad (p \neq q)$ sub into (1) $x = 2cp - p^2 \left(\frac{2cq}{p + q} \right)$ $= \frac{2cp^2 + 2cpq - 2cp^2}{p + q}$ $= \frac{2cpq}{p + q}$ $\therefore R \text{ is } \left(\frac{2cpq}{p + q}, \frac{2c}{p + q} \right)$	2 marks : correct solution 1 mark : significant progress towards correct solution

(a)(iii)

$$x = \frac{2cpq}{p+q} \quad \dots(1)$$

$$y = \frac{2c}{p+q} \quad \dots(2)$$

$$(1) \div (2) \quad \frac{x}{y} = pq$$

$$\text{and from (2)} \quad p+q = \frac{2c}{y}$$

$$\text{Now, given that } p^2 + q^2 = 2$$

$$(p+q)^2 - 2pq = 2$$

$$\left(\frac{2c}{y}\right)^2 - 2\left(\frac{x}{y}\right) = 2$$

$$4c^2 - 2xy = 2y^2$$

$$xy + y^2 = 2c^2$$

OR sub $R\left(\frac{2cpq}{p+q}, \frac{2c}{p+q}\right)$ into $xy + y^2 = 2c^2$ to show true

(b)

S lies on PQ

$$\text{so } m_{PS} = m_{QS}$$

$$\frac{b \tan \theta - 0}{a \sec \theta - ae} = \frac{b \tan \phi - 0}{a \sec \phi - ae}$$

$$\frac{\tan \theta}{\sec \theta - e} = \frac{\tan \phi}{\sec \phi - e}$$

$$\tan \theta \sec \phi - e \tan \theta = \tan \phi \sec \theta - e \tan \phi$$

$$e(\tan \theta - \tan \phi) = \tan \theta \sec \phi - \tan \phi \sec \theta$$

$$e = \frac{\tan \theta \sec \phi - \tan \phi \sec \theta}{\tan \theta - \tan \phi} \times \frac{\cos \theta \cos \phi}{\cos \theta \cos \phi}$$

$$= \frac{\sin \theta - \sin \phi}{\sin \theta \cos \phi - \cos \theta \sin \phi}$$

$$= \frac{\sin \theta - \sin \phi}{\sin(\theta - \phi)}$$

3 marks : correct solution

2 marks : substantial progress towards correct solution

1 mark : partial progress towards correct solution

4 marks : correct solution

3 marks : substantial progress towards correct solution

2 marks : partial progress towards correct solution

1 mark : recognising that $m_{PS} = m_{QS}$

(c)(i)(A)	$\frac{x^2}{4-\lambda} + \frac{y^2}{2-\lambda} = 1$ for ellipse, $4-\lambda > 0$ and $2-\lambda > 0$ $\lambda < 4$ and $\lambda < 2$ ie <u>$\lambda < 2$</u>	<u>1 mark</u> : correct solution
(c)(i)(B)	for hyperbola, $4-\lambda > 0$ and $2-\lambda < 0$ $\lambda < 4$ and $\lambda > 2$ ie $2 < \lambda < 4$ OR for hyperbola, $4-\lambda < 0$ and $2-\lambda > 0$ $\lambda > 4$ and $\lambda < 2$ ie no solution $\therefore \underline{2 < \lambda < 4}$	<u>1 mark</u> : correct solution
(c)(ii)	As λ increases from 1 to 2, the curve <u>remains as an ellipse</u>, with both the major and minor axes decreasing. The ellipse <u>becomes flatter</u>, or more ‘cigar’ shaped. As λ approaches 2, the ellipse approaches a <u>line segment</u> joining $(-\sqrt{2}, 0)$ to $(\sqrt{2}, 0)$.	<u>3 marks</u> : correct and complete explanation <u>2 marks</u> : reasonably correct explanation <u>1 mark</u> : partially correct